

SMALL-PARAMETER COUNTEREXAMPLES TO CONJECTURE 3 OF LARSSON, MANABE, AND MIYADERA

ABSTRACT. Larsson, Manabe, and Miyadera conjectured that three parameterized families of subtraction Nim with a one-time pass satisfy several properties of the Grundy sequence, among them two eventual mex identities. Under the printed quantifiers $a, n \in \mathbb{N}$ in their Conjecture 3, every positive integer a yields a counterexample in each subpart: the exceptional parameters we exhibit are respectively $n = 2$, $n = 1$ and $n = 2$. None of the three witnesses is covered by the theorems they prove, which treat only the subtraction set $\{2, 4n, 4n + 2\}$ with $n \geq 3$.

1. STATEMENT AND NOTATION

Let $S \subset \mathbb{N}$ be a subtraction set. Write $\mathcal{G}_S(x, 0)$ for the Grundy value with no pass available and $\mathcal{G}_S(x, 1)$ for the value with a one-time pass available. Following the recursion used in [1],

- (1) $\mathcal{G}_S(x, 0) = \text{mex}\{\mathcal{G}_S(x - s, 0) : s \in S, s \leq x\},$
- (2) $\mathcal{G}_S(x, 1) = \text{mex}(\{\mathcal{G}_S(x - s, 1) : s \in S, s \leq x\} \cup \{\mathcal{G}_S(x, 0)\}),$

whenever $x > 0$ admits a subtraction move, and $\mathcal{G}_S(0, 1) = 0$. All positive positions used below admit the smallest subtraction, so no convention concerning passes from positive terminal positions enters the argument.

Conjecture 3 of [1] concerns three parameterized subtraction sets $S = \{s_1, s_2, s_3\}$ and the three properties displayed there as Equations (35), (36) and (37), which we reproduce in that order:

- (3) $\mathcal{G}_S(x, 0) = \text{mex}\{\mathcal{G}_S(x + s, 0) : s \in S\},$
- (4) $\mathcal{G}_S(x, 1) = \text{mex}\{\mathcal{G}_S(x - s, 1) : s \in S\},$
- (5) $\mathcal{G}_S(x, 1) = \text{mex}\{\mathcal{G}_S(x + s, 1) : s \in S\}.$

Its three subparts, labelled there (a), (b) and, in place of (c), “(iii)”, are each quantified over $a, n \in \mathbb{N}$, and each is a conjunction:

- (a) for $S = \{a, 2an, 2an + a\}$, that (3) holds, that (4) and (5) hold for $x \geq 6an + (4a + 1)$, and that the sequence of Grundy numbers enters into a loop when $x \geq 6an + (4a + 1)$;

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- (b) for $S = \{a, (2n+1)a, (2n+3)a\}$, that (3) holds, that (4) and (5) hold for $x \geq a(2n+3) + 1$, that $\mathcal{G}_S(x, 1) < 4$ for every x , and that the sequence enters into a loop when $x \geq a(2n+3) + 1$;
- (iii) for $S = \{a, (2n+1)a, (2n+5)a\}$, that (3) holds, that $\mathcal{G}_S(x, 1) < 4$ for every x , that (4) and (5) hold for $x \geq a(2n+5) + 1$, and that the sequence enters into a loop when $x \geq a(2n+5) + 1$.

No restriction on n beyond $n \in \mathbb{N}$ appears in the conjecture.

Equation (5) is the intended form of Equation (37) of [1], whose displayed left-hand side reads $\mathcal{G}(x)$ and so omits the second argument. The sentence introducing that display reads “(iii) The sequence of Grundy numbers $\mathcal{G}(x, 1)$ has the reverse-mex property”, which supplies the argument 1. Reading the omitted argument as 0, and with it the arguments printed as 1 on the right of Equation (37), would turn that equation into Equation (35), reproduced here as (3), which each subpart already asserts separately and with no threshold. The remaining, mixed reading, with $\mathcal{G}(x, 0)$ on the left and the printed argument 1 on the right, also fails at the witness of part (a): at $a = 1$ and $n = 2$ the table in the proof of Theorem 3 gives $\text{mex}\{2, 1, 0\} = 3$ for the right-hand side at $x = 18$, against a left-hand side of 0. The failure of (4) recorded below does not depend on this reading, as Equation (36) is printed with both arguments throughout.

Since each subpart is a conjunction, it is refuted by refuting any one conjunct; Theorem 3 below refutes each subpart through the conjunct (4), and part (a) also through (5). The other conjuncts are not contradicted at the witnesses exhibited: (3) holds at each of them, as the table in the proof of Theorem 3 shows for part (a) and as the parity formula of Lemma 2 shows, through Lemma 1, for the other two; and the bound $\mathcal{G}_S(x, 1) < 4$ asserted in parts (b) and (iii) holds at every x for the odd sets covered by Lemma 2, which give $\mathcal{G}_S(x, 1) \leq 2$. Nothing is asserted here about the eventual periodicity claimed alongside.

Lemma 1 (Scaling). *Suppose $1 \in S$ and $a \in \mathbb{N}$. Then, for every $k \in \mathbb{Z}_{\geq 0}$ and $p \in \{0, 1\}$,*

$$\mathcal{G}_{aS}(ak, p) = \mathcal{G}_S(k, p), \quad aS := \{as : s \in S\}.$$

Proof. Induct on k , proving the case $p = 0$ before the case $p = 1$ at each k . For $p = 0$, every option from ak is of the form $a(k - s)$, so (1) and the induction hypothesis give the claim. Having established the no-pass value at ak , Equation (2) gives the with-pass value from the lower with-pass values and the current no-pass value. These option sets coincide with those at k for S . The initial values at $k = 0$ are both zero, and $1 \in S$ ensures that every positive position ak admits the subtraction a . $\square$

Lemma 2 (Odd subtraction sets). *Let S consist entirely of odd integers and contain 1. Then*

$$\mathcal{G}_S(x, 0) = \begin{cases} 0, & x \text{ even}, \\ 1, & x \text{ odd}, \end{cases}$$

and

$$\mathcal{G}_S(x, 1) = \begin{cases} 0, & x = 0, \\ 1, & x > 0 \text{ even}, \\ 2, & x \text{ odd and } x \in S, \\ 0, & x \text{ odd and } x \notin S. \end{cases}$$

Proof. Both formulas are established by a single simultaneous strong induction on x , the parity formula for $\mathcal{G}_S(x, 0)$ being proved first at each x , since the with-pass value at x uses the no-pass value at x . The first formula follows because every legal move changes parity. For $x > 0$ even, all subtraction options have odd heap size, so their with-pass values are 0 or 2 by induction, while the pass contributes $\mathcal{G}_S(x, 0) = 0$; hence the mex is 1.

For x odd, every subtraction option has even heap size. If $x \in S$, one option is 0, while every other legal subtraction option and the pass have value 1, so the mex is 2. If $x \notin S$, all subtraction options and the pass have value 1, so the mex is 0. $\square$

2. COUNTEREXAMPLES

Theorem 3. *Under the printed quantifiers $a, n \in \mathbb{N}$, every subpart of Conjecture 3 in [1] is false. More precisely, for every $a \in \mathbb{N}$:*

- (1) *In part (a), take $n = 2$, so $S = a\{1, 4, 5\}$. At $x = 18a$, both (4) and (5) fail.*
- (2) *In part (b), take $n = 1$, so $S = a\{1, 3, 5\}$. At $x = 6a$, Equation (4) fails.*
- (3) *In the third subpart, printed as “(iii)”, take $n = 2$, so $S = a\{1, 5, 9\}$. At $x = 10a$, Equation (4) fails.*

Proof. By Lemma 1, it is enough to compute at scale $a = 1$.

For $T = \{1, 4, 5\}$, successive use of (1) and (2) gives

x	0	1	2	3	4	5	6	7	8	9	10	11
$\mathcal{G}_T(x, 0)$	0	1	0	1	2	3	2	3	0	1	0	1
$\mathcal{G}_T(x, 1)$	0	2	1	0	1	4	0	2	3	0	1	3

and

x	12	13	14	15	16	17	18	19	20	21	22	23
$\mathcal{G}_T(x, 0)$	2	3	2	3	0	1	0	1	2	3	2	3
$\mathcal{G}_T(x, 1)$	0	1	3	0	1	2	4	2	3	0	1	0

Consequently,

$$\text{mex}\{\mathcal{G}_T(17, 1), \mathcal{G}_T(14, 1), \mathcal{G}_T(13, 1)\} = \text{mex}\{2, 3, 1\} = 0 \neq 4 = \mathcal{G}_T(18, 1),$$

and

$$\text{mex}\{\mathcal{G}_T(19, 1), \mathcal{G}_T(22, 1), \mathcal{G}_T(23, 1)\} = \text{mex}\{2, 1, 0\} = 3 \neq 4 = \mathcal{G}_T(18, 1).$$

Thus both required identities fail in part (a).

For $S = \{1, 3, 5\}$, Lemma 2 gives

$$\mathcal{G}_S(1, 1) = \mathcal{G}_S(3, 1) = \mathcal{G}_S(5, 1) = 2, \quad \mathcal{G}_S(6, 1) = 1.$$

Hence

$$\text{mex}\{\mathcal{G}_S(5, 1), \mathcal{G}_S(3, 1), \mathcal{G}_S(1, 1)\} = \text{mex}\{2, 2, 2\} = 0 \neq 1 = \mathcal{G}_S(6, 1).$$

This disproves part (b).

For $U = \{1, 5, 9\}$, Lemma 2 similarly gives

$$\mathcal{G}_U(1, 1) = \mathcal{G}_U(5, 1) = \mathcal{G}_U(9, 1) = 2, \quad \mathcal{G}_U(10, 1) = 1,$$

so

$$\text{mex}\{\mathcal{G}_U(9, 1), \mathcal{G}_U(5, 1), \mathcal{G}_U(1, 1)\} = \text{mex}\{2, 2, 2\} = 0 \neq 1 = \mathcal{G}_U(10, 1).$$

This disproves the third subpart.

Finally, the conjectured thresholds specialize at the chosen parameters: $6an + (4a + 1) = 16a + 1$ at $n = 2$ in part (a), $a(2n + 3) + 1 = 5a + 1$ at $n = 1$ in part (b), and $a(2n + 5) + 1 = 9a + 1$ at $n = 2$ in the third subpart. The witnesses satisfy

$$18a \geq 16a + 1, \quad 6a \geq 5a + 1, \quad 10a \geq 9a + 1,$$

for every $a \geq 1$. In part (b) and in the third subpart the witness is $x = \min S + \max S$, which exceeds the corresponding threshold by exactly $a - 1$; the two witnesses therefore meet their thresholds with equality at $a = 1$. $\square$

Remark 4 (Scope). The abstract and the principal definitions of [1] restrict the family $\{2, 4n, 4n+2\}$ to $n \geq 3$, the abstract adding: “We do not treat the case that $n = 1$ or $n = 2$ in this article.” Conjecture 3, however, separately quantifies $a, n \in \mathbb{N}$ in each subpart, and Theorem 3 addresses that printed statement. None of the three witnesses lies in the family treated by the theorems of [1]. In part (a) the set $\{a, 2an, 2an+a\}$ has the form $\{2, 4m, 4m+2\}$ only when $a = 2$, and there the choice $n = 2$ gives $\{2, 8, 10\}$, one of the two cases explicitly set aside. In part (b) and in the third subpart no member of the family can have that form: matching the least elements forces $a = 2$, and the middle element is then $4n + 2 \equiv 2 \pmod{4}$, whereas $4m \equiv 0 \pmod{4}$. Accordingly Theorem 3 does not contradict the proved $n \geq 3$ results in [1], and it makes no assertion about Conjecture 3 after imposing the additional restriction $n \geq 3$. The parameters of Theorem 3 are not claimed to be the only exceptional ones. For $a = 1$, where $1 \in S$ and no positive position is terminal, direct computation for $n = 1, \dots, 12$ finds no others: (4) or (5) fails above the stated threshold at $n = 2$, $n = 1$ and $n = 2$ respectively, and at no other n in that range. For $a \geq 2$ each family contains positions from which no subtraction is legal, and the set of exceptional n then depends on the convention adopted for the pass at such positions. Theorem 3 is independent of that convention, since every position it uses admits the subtraction a .

REFERENCES

- [1] U. Larsson, H. Manabe, and R. Miyadera, *A subtraction Nim with a pass*, arXiv:2605.14321v1 [math.CO] (2026), arXiv:2605.14321.